

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

Software

The software includes Adafruit library for DHT sensors, virtual wire library for communicating with 433MHz RF sensors and liquid crystal library for the LCD display. All libraries are included in the software bundle in this month's DVD.

Construction and testing

An actual-size, single-side PCB of the transmitter unit is shown in Fig. 4 and its component layout in Fig. 5. An actual-size, single-side PCB of the receiver unit is shown in Fig. 6 and its component layout in Fig. 7.

Connect the battery of the transmitter and then place the transmitter at a suitable location inside the fridge. Close the fridge door and power on the receiver unit. Temperature and RH will be shown on the LCD. Move the receiver unit to the farthest corner of the house and, in most likelihood, you will still be able to see the temperature and RH ticking. **EFY**

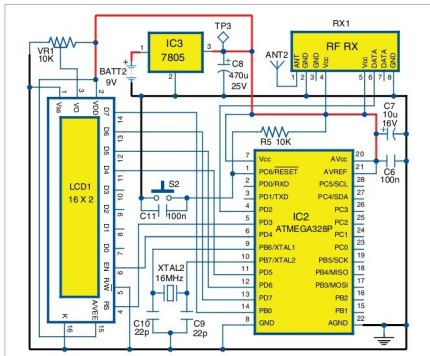


Fig. 3: Circuit diagram of the receiver unit

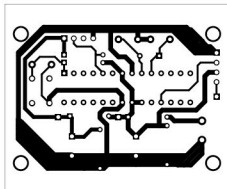


Fig. 4: Actual-size PCB pattern of the transmitter unit

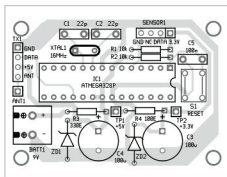


Fig. 5: Component layout of the PCB shown in Fig. 4

signal is valid. To circumvent this problem, a counter has been provided on the left side of the LCD

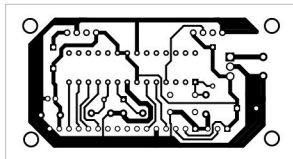


Fig. 6: Actual-size PCB pattern of the receiver unit

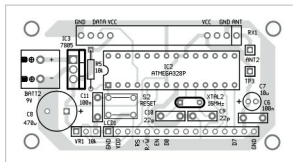


Fig. 7: Component layout of the PCB shown in Fig. 6

display. If the counter does not move, or stops, it means that the incoming signal has stalled. Reset the system by pressing switch S2 momentarily to come back to normal operation.



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